

M.Tech Part I/PhD Mining Engineering
Mid-Term Examination, 2024
MN573: Safety & Risk Management in Mines

Time: 2 Hrs.

Marks: 25

1. Discuss how Heinrich's Safety Philosophy is a phenomenal departure from the then prevailing safety concept. Using the concept of 'Domino Game' elaborate the injury sequence of mine accidents and their preventive measures. (5)
2. Explain the process of risk analysis briefly. How does qualitative risk analysis differ from quantitative risk analysis? Explain the process of developing a risk matrix. (5)
3. Outline the salient features of The Occupational Safety, Health And Working Conditions Code, 2020. (5)
4. Calculate the mean and standard deviation of binomial random variable. (5)
5. Do as directed:
 - (i) A loader while loading coal fell down due to the slippery floor. As a consequence, he received a fracture in his tibia bone. Identify the type of injury. (1)
 - (ii) A mason was constructing the permanent stopping wall with bricks to regulate the air which was located near the coal face. Due to blast vibration, one brick was fallen on his head, and he suffered with an injury in his skull. Identify possible unsafe act(s) and/or unsafe condition(s). (1)
 - (iii) Distinguish between hazard and risk. (1)
 - (iv) A person has lost one of his eyes in an occupational accident. As per ASTC, how many days are charged for this occupational injury? (1)
 - (v) In an underground coal mine, the accident statistics for the year 2022 revealed that there were one death, four serious bodily injuries and 18 reportable injuries. The coal mine operates 310 days in a year, three shifts in a day and eight hours in a shift. The average employment in a shift was 400 and the production of the mine was 0.9 million tonne. Calculate fatality rate per thousand persons employed. (1)

2. Describe the concept of Integrated Marketing Communications (IMC) and explain how it can be effectively implemented using the AIDA (Attention, Interest, Desire, Action) model. Provide examples of how different communication channels can be used at each stage of the AIDA model to create a cohesive brand message. How does this integrated approach benefit both the brand and the consumer?
3. Discuss the importance of personal branding in the digital age and how it relates to professional success. Explain the key elements of a strong personal brand and provide strategies for developing and maintaining one's personal brand online. How can individuals measure the effectiveness of their personal branding efforts? Consider potential challenges and ethical considerations in personal branding.
4. Analyze the impact of social media on consumer behavior and brand loyalty. How have social media platforms changed the way consumers interact with brands and make purchasing decisions? Discuss strategies that brands can use to leverage social media for building and maintaining customer loyalty. Include considerations of both the benefits and potential risks of relying on social media for brand building.